MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHYA01 - STELLAR ASTROPHYSICS- RADIATIVE PROCESS, STELLAR STRUCTURE AND EVOLUTION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **3** marks

1. Write the differential equation describing absorption of radiation in a gas. [K] / [CO2]

Ans:The change in intensity I of radiation is given by: $dI = -\kappa\rho I\, ds$

ρ=density, I = intensity of radiation ,s = path length, κ = opacity

or $dI_\lambda = -\alpha_\lambda I_\lambda\, ds$

α_λ= absorption coefficient

Equation only 2 marks. Terms explained-1 mark

2. Write the relation between energy density and specific intensity. [K] / [CO1]

Ans: The energy density u is related to the specific intensity I by:

$u = 4\pi I/c$

u = energy density , I = specific intensity, c = speed of light

or $u_\nu = \int u_\nu(\Omega)\, d\Omega = (1/c)\int I_\nu\, d\Omega = 4\pi J/c$

$u = \int u_\nu\, d\nu = (4\pi/c)\int J_\nu\, d\nu$

Relation only give 2 marks. Terms explained-1 mark

3. Write the mathematical form of Stefan–Boltzmann law. [K] / [CO2]

Ans: $F = \sigma T^4$

F = total flux emitted per unit area, σ = Stefan–Boltzmann constant, T = absolute temperature

or $L = 4\pi r^2 \sigma T^4$ L= luminosity

or $U(T) = aT^4$ U(T) = energy density

Equation only 2 marks. Terms explained-1 mark

4. Give one astrophysical situation where line broadening is significant. [A] / [CO3]

Ans: Line broadening is significant in stellar atmospheres, where it is used to determine physical properties such as temperature, pressure, and density of stars. **Or**

Spectral line broadening is defined as the phenomenon in which a spectral line has a finite width instead of being infinitely sharp, due to various physical processes such as natural broadening, Doppler broadening, and pressure (collision) broadening.Line broadening is significant in stellar atmospheres.

Line broadening explanation only-1 mark . Mention natural broadening and doppler broadening - 2 marks.Any one astrophysical situation give 3marks

5. Explain carbon and oxygen burning in stars. [U] / [CO4]

Ans:if both carbon and oxygen burning are explained briefly,give 2 marks. If only one type is explained give 1 mark

6. Explain briefly the luminosity classification (MK system) of stars. [U] / [CO4]

Ans:2 marks for writing the classification.

I	Supergiants
II	Bright giants
III	Giants
IV	Subgiants
V	Main-sequence (dwarfs)

luminosity class is based on line widths and surface gravity.(1 mark if only this)

7. Discuss, with reference to the Virial theorem,why a contracting star becomes hotter

Ans.For a star of mass M and radius R:

$U \approx -\frac{GM^2}{R}$ As the star contracts (R decreases), U becomes more negative

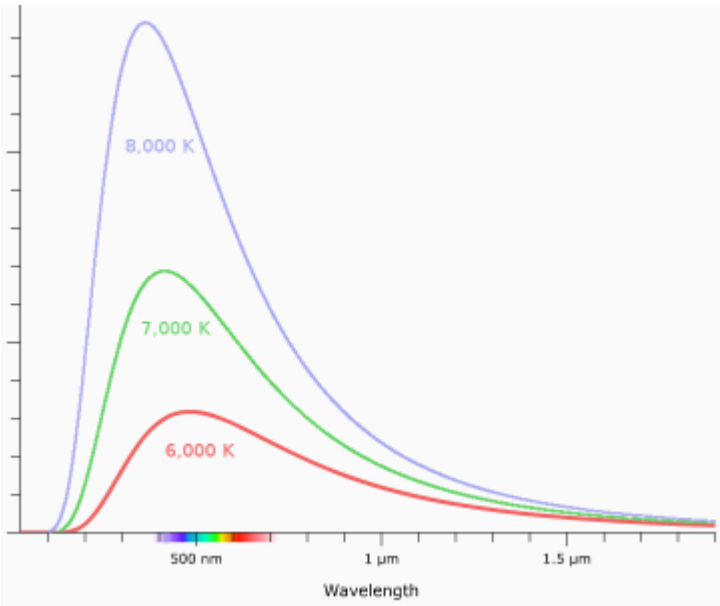
From the virial relation: $K = -\frac{1}{2}U$

If U becomes more negative, then K must increase. As a result, T increases.

(Give 2 marks if the explanation is given clearly. Give one mark if virial theorem is stated or partially explained the reason for rise in temperature in contraction.)

8.Explain the origin and characteristics of stellar spectra. How do absorption lines arise in the spectrum of a star?

Ans. stellar spectrum represents the distribution of em radiation emitted by stars across wavelengths which approximates black body radiation. OR



(Give 1 mark.)

When the photons from the interior of a star travel outward,atoms in its outer atmosphere absorb

photons of specific energies corresponding to atomic transitions. As a result dark absorption lines arise in the observed stellar spectrum. - (give 1 mark)

9. Define dynamical (free-fall) timescale. [K] / [CO4]

Ans.

Time scale on which a star would expand or contract in case of departure from the condition of hydrostatic equilibrium (free-fall time scale) /The time required for a star to collapse under its own gravity if the internal pressure support were suddenly removed.

$$\tau_{dyn} \equiv \frac{\text{characteristic radius}}{\text{characteristic velocity}} = \frac{R}{v_{esc}}$$

$$\tau_{dyn} = \sqrt{\frac{R^3}{2GM}} \approx \text{(for Sun) } 1100 \text{ s}$$

$$\tau_{dyn} << \tau_{KH} << \tau_{nuc}$$

Most stars, most of the time, are in hydrostatic and thermal equilibrium, with slow changes in structure and composition occurring on the (long) time scale

(Any one of the definition with equation or characteristics of dynamic time scale give full marks)

10. What is the 21-cm line? [U] / [CO4

Ans.

1. The 21-cm line arises from a **spin-flip transition** in the hydrogen atom. It corresponds to the **Wavelength** = 21.1 cm and **Frequency** = 1420 MHz
2. Hydrogen atoms consist of a proton and electron, possessing Intrinsic spin angular momentum. They are charged particles, their spins create **magnetic dipole moments**.The two possible configurations are **Aligned spins (parallel)** → slightly higher energy & **Anti-aligned spins (antiparallel)** → lower energy. When the electron flips from aligned to anti-aligned —--Energy is released—A **21-cm photon is emitted**.
3. Application in astronomy are It reveals the structure of the interstellar medium (ISM), Maps neutral hydrogen,Reveals galactic structure, Measures velocities via Doppler shift, Estimates gas mass, Penetrates dust, Helps study the early universe

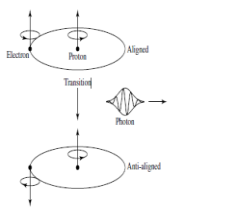


FIGURE 5 When the spins of the electron and proton in a hydrogen atom go from being aligned to being anti-aligned, a 21-cm-wavelength photon is emitted.

(First Point with any explanation or diagramme give full marks)

11. On what does Eddington luminosity depend? [U] / [CO5]

Ans.

The **Eddington luminosity** is the maximum luminosity a star (or any radiating object) can have such that the **outward radiation pressure balances the inward gravitational force**. Beyond this limit, radiation can push matter away from the star.

$$L_{\mathrm{Edd}} = \frac{4\pi G M m_{\mathrm{p}} c}{\sigma_{\mathrm{T}}}$$

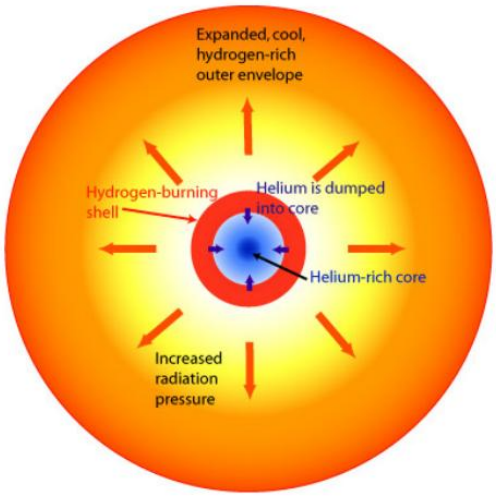
$$\cong 1.26 \times 10^{31} \left(\frac{M}{M_{\odot}}\right) \mathrm{W} = 1.26 \times 10^{38} \left(\frac{M}{M_{\odot}}\right) \mathrm{erg/s} = 3.2 \times 10^{4} \left(\frac{M}{M_{\odot}}\right) L_{\odot}$$

12. What happens to the core after hydrogen exhaustion? [K] / [CO5]

Ans.

The RGB phase begins after a star exhausts hydrogen in its core. Without central nuclear fusion, the core contracts, while the surrounding layers heat up and ignite **hydrogen shell Burning (** As the core contracts, the star's envelope expands and cools. This causes the star to move toward the "red" (cooler) and more "luminous" part of the H-R diagram, eventually reaching the **Hayashi line**

Luminosity increases sharply while the effective temperature remains approximately constant



(Explanation or diagramme give full mark) **[3x10 = 30]**

Part B

Short Essay/ Problems

Answer any **4** questions

Each question carries **5** marks

13. Describe the inverse square law. [U] / [CO1]

Ans. An isotropic source emits energy equally in all directions. Imagine two spheres, S_1 and S with radius r_1 and r near the source. By conservation of energy, the total energy passing through S_1 must be the same as that passing through S.

The energy radiated ,$dE = F\, dA\, dt$.

$F(r_1) \cdot 4\pi r_1^2 = F(r) \cdot 4\pi r^2$

If sphere S_1 is fixed:

$F(r) = \text{constant}/r^2$

$\Rightarrow F(r) \propto 1/r^2$

The radiative flux from a source decreases inversely as the square of the distance from the source.This is the Inverse Square Law.

Give 5 marks to full explanation.Law / Equation only-2 marks.

14. Describe how the expression for mean intensity is obtained under equilibrium

Ans.Under thermodynamic equilibrium, the specific intensity I_v is exactly equal to the Planck function $B_v(T)$.Since the field is isotropic in equilibrium:

$I_v = B_v(T)$ ……..1

$\int I_v \, d\Omega = B_v(T) \int d\Omega = 4\pi B_v(T)$ …….2

Substituting this into the mean intensity formula:

$J_v = (1 / 4\pi) (4\pi B_v(T)) = B_v(T)$.....3

Equation 1 only - 2marks. Complete steps- 5 marks. Mean intensity $J_v = 1/4\pi \int I_v d\Omega$ -2marks

15. Discuss the concept of (a) stellar mass and (b) stellar radius.

Ans.a)stellar mass (2.5 mark)

Explaining term stellar mass and the classification -high mass star,low mass star, sun like star

OR explaining stellar mass calculation using Kepler's 3rd law : $P^2 = 4\boldsymbol{\pi}^2 \ a^3/G(m1 + m2)$

(give 2.5 mark).

b) stellar radius.(2.5 mark)

Refers to the distance from a star's centre to to its outer surface.(give 1 mark)

If either

Direct measurement method ($R = d \ \theta/2$)

Or indirect measurement method ($R = (L/4\boldsymbol{\pi\sigma}T^4)^{1/2}$.

Or written both methods without equations give 1.5 mark

16. Give a detailed account of stellar structure and the fundamental equations governing stellar equilibrium and energy transport.

Ans.

stellar structure refers to the **radial variation of physical quantities** inside a star, such as:

- Pressure $P(r)$
- Density $\rho(r)$

- Temperature $T(r)$
- Luminosity $L(r)$
- Enclosed mass $M(r)$

(2 mark)

Hydrostatic equilibrium

Hydrostatic equilibrium is the **primary condition for stellar stability**.

$$\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2}$$

This equation states that **inside a star, inward gravity is exactly balanced by the outward pressure gradient**.

(1.5 mark if either equation with terms explained or the statement is given)

Virial Theorem

The Virial Theorem for a Star in Equilibrium

For a star in hydrostatic equilibrium:

2K+U=0

where

- K= total internal kinetic energy (thermal energy of gas particles)
- U= total gravitational potential energy

(1.5 mark if either equation with terms explained or the statement is given)

17. Compare white dwarfs and neutron stars. [A] / [CO5]

Ans.

Neutron Star	Blackhole
Neutron Star mass range– 1.4 to 3 solar mass	Mass range - > 3 times of solar mass
Mostly Neutrons	A region of spacetime, not a solid object
Emit radiation/light	Does not emit radiation, light from accretion disk
Rotating neutron stars are Pulsars	Parts of blackholes are singularity, event horizon etc.

Less dense than blackhole	Most dense

(Any relevant 5 points give full credits)

18. Discuss formation and evolution of GMCs. [U] / [CO4]

Ans :-

- Dense regions of gas and dust in the **interstellar medium (ISM)** where physical conditions such as temperature, density, and mass vary widely known as Interstellar clouds. They are Classified into Diffuse / Translucent molecular Clouds & Giant Molecular Clouds (GMCs). Giant Molecular Clouds are the regions of star formation. The regions of gas and dust in the ISM are Molecular Clouds.The main characteristics of GMC are Large size with typical diameters 50pc,Cloud collapses and stars form due to gravity, Temperature: ~15 K, Number density: 1×10^{8} to 3×10^{8} m^{-33} (Density 10^{20} times of a typical star), Hundreds of thousands of times heavier than the Sun (10^{5}), Size: ~50 pc
- A cloud begins to collapse when **gravity overcomes internal pressure**.This condition is described by the **Jeans Mass**: If $M_{cloud}>M_{J}$. When the collapsing GMC attains high density, it becomes gravitationally unstable and breaks up into smaller pieces.When fragmentation happens the turbulence inside the cloud, Shock waves from nearby supernovae, Galactic rotation, Magnetic field effects. Each fragment can form A single star, A binary star system,A small star clusterThe fragmentation of collapsing clouds leads to the formation of protostars − pre-stellar objects which are hot enough to radiate infrared radiation but not hot enough to generate energy by nuclear fusion.
- Fragmentation of collapsing cloud leads to a stage when a protostar is formed and its interior attains hydrostatic equilibrium.
- The interior of the protostar at this stage becomes almost convective
- The stellar object so formed is called a pre-main sequence star.

  (Definition of GMC with any three properties give full marks)